End-to-End Inference Procedure

Neural-Network Subgrid Flux Modeling + MCL for the 1-D Shallow-Water Equations

The one idea to remember

At inference time, the neural network does *not* predict the fine-grid solution. It predicts a *subgrid flux correction*. That correction is inserted into a coarse finite-volume solver, and MCL limits the correction when necessary so that the intermediate states remain physically admissible.

coarse state → LLF flux + NN subgrid flux → MCL → coarse PDE update → next coarse state

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1 1. The underlying shallow-water problem

The paper considers the one-dimensional shallow-water equations

$$\frac{\partial}{\partial t}\begin{bmatrix} h \\ q \end{bmatrix} + \frac{\partial}{\partial x}\begin{bmatrix} q \\ \dfrac{q^2}{h} + \dfrac{1}{2}gh^2 \end{bmatrix} = 0.$$

The conserved state is

$$\mathbf{u} = \begin{bmatrix} h \\ q \end{bmatrix}, \qquad q = hv,$$

and the physical flux is

$$\mathbf{f}(\mathbf{u}) = \begin{bmatrix} q \\ \dfrac{q^2}{h} + \dfrac{1}{2}gh^2 \end{bmatrix}.$$

Symbol	Meaning
h	water height/depth
v	fluid velocity
$q = hv$	discharge
g	gravitational acceleration
$\mathbf{u} = [h, q]^T$	conserved state
$\mathbf{f}(\mathbf{u})$	physical flux

2 2. Fine grid and coarse grid

The fine simulation has N_f cells of width Δx. The coarse grid groups k fine cells into one macrocell:

$$N_c = \frac{N_f}{k}, \qquad \Delta X = k\Delta x.$$

Fine states:

$$\mathbf{u}_1, \ldots, \mathbf{u}_{N_f}.$$

Coarse states:

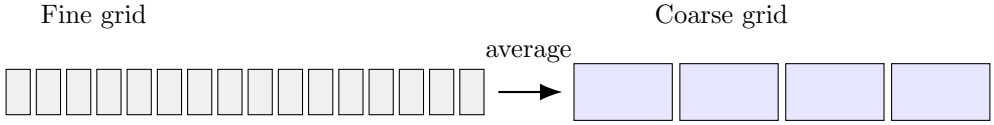
$$\boxed{\mathbf{U}_i = \frac{1}{k}\sum_{j=k(i-1)+1}^{ki} \mathbf{u}_j.}$$

Thus the coarse model stores only

$$\mathbf{U}_1, \ldots, \mathbf{U}_{N_c}.$$

Important

The reduced model does *not* reconstruct the individual fine-grid values during inference. Its target is the evolution of the coarse averages.



3 3. Coarse LLF flux and the missing subgrid flux

At interface $i+\frac{1}{2}$, the coarse LLF flux is

$$\boxed{\bar{\mathbf{F}}_{i+\frac{1}{2}} = \mathcal{F}(\mathbf{U}_i, \mathbf{U}_{i+1}) = \frac{\mathbf{f}(\mathbf{U}_i) + \mathbf{f}(\mathbf{U}_{i+1})}{2} - \frac{\Lambda_{i+\frac{1}{2}}}{2}(\mathbf{U}_{i+1} - \mathbf{U}_i).}$$

The wave-speed bound is

$$\boxed{\Lambda_{i+\frac{1}{2}} = \max\left(|v_i| + \sqrt{gh_i}, |v_{i+1}| + \sqrt{gh_{i+1}}\right).}$$

The true fine-grid flux at the same physical interface is

$$\mathbf{F}_{ki+\frac{1}{2}} = \mathcal{F}(\mathbf{u}_{ki}, \mathbf{u}_{ki+1}).$$

The paper defines the subgrid flux correction as

$$\boxed{\mathbf{G}_{i+\frac{1}{2}} = \mathbf{F}_{ki+\frac{1}{2}} - \bar{\mathbf{F}}_{i+\frac{1}{2}}.}$$

Therefore the desired effective coarse flux is

$$\boxed{\mathbf{F}^{\text{effective}}_{i+\frac{1}{2}} = \bar{\mathbf{F}}_{i+\frac{1}{2}} + \mathbf{G}_{i+\frac{1}{2}}.}$$

The coarse LLF flux is the baseline; $\mathbf{G}$ is the missing effect of the unresolved fine scales.

4 4. What the neural network learns

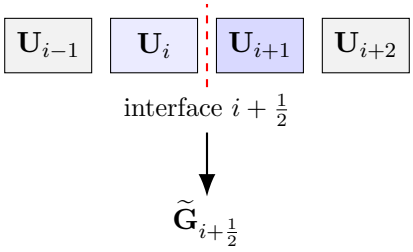
The paper uses the four-point coarse stencil

$$\boxed{(\mathbf{U}_{i-1}, \mathbf{U}_i, \mathbf{U}_{i+1}, \mathbf{U}_{i+2}).}$$

The network outputs

$$\boxed{\widetilde{\mathbf{G}}_{i+\frac{1}{2}} = \widetilde{\mathcal{G}}(\mathbf{U}_{i-1}, \mathbf{U}_i, \mathbf{U}_{i+1}, \mathbf{U}_{i+2}; \omega) = \begin{bmatrix} \widetilde{\mathbf{G}}^h_{i+\frac{1}{2}} \\ \widetilde{\mathbf{G}}^q_{i+\frac{1}{2}} \end{bmatrix}.}$$

Here ω denotes the learned weights and biases.



The four-point stencil is used because the authors found that a two-point stencil does not provide enough coarse-scale information for accurate subgrid-flux modeling.

5 5. Training versus inference

5.1 Training

During training, the expensive fine simulation is available. The authors generate samples

$$\left(\mathbf{U}_{i-1}^{(j)}, \mathbf{U}_{i}^{(j)}, \mathbf{U}_{i+1}^{(j)}, \mathbf{U}_{i+2}^{(j)}, \mathbf{G}_{i+\frac{1}{2}}^{(j)}\right).$$

The NN is trained using

$$\boxed{\mathcal{C}(\omega) = \frac{1}{M}\sum_{j=1}^{M}\left\|\widetilde{\mathcal{G}}(\mathbf{U}_{i-1}^{(j)}, \mathbf{U}_{i}^{(j)}, \mathbf{U}_{i+1}^{(j)}, \mathbf{U}_{i+2}^{(j)}; \omega) - \mathbf{G}_{i+\frac{1}{2}}^{(j)}\right\|_2^2.}$$

5.2 Inference

After training, the fine-grid states are *not* available or needed. At inference time, only

$$\mathbf{U}_1, \ldots, \mathbf{U}_{N_c}$$

are evolved.

Training/inference distinction

Training:

fine solution $\rightarrow$ true G $\rightarrow$ train NN.

Inference:

coarse solution $\rightarrow$ NN predicts G $\rightarrow$ coarse solver.

6 6. Inference: start from a coarse state

At coarse time t^n, suppose we know

$$\boxed{\mathbf{U}_1^n, \ldots, \mathbf{U}_{N_c}^n.}$$

We want

$$\mathbf{U}_1^{n+1}, \ldots, \mathbf{U}_{N_c}^{n+1}.$$

For every interface $i + \frac{1}{2}$, perform the following steps.

7 7. Step 1 — compute LLF quantities

From $\mathbf{U}_i$ and $\mathbf{U}_{i+1}$ compute:

1. the physical fluxes $\mathbf{f}(\mathbf{U}_i)$ and $\mathbf{f}(\mathbf{U}_{i+1})$;
2. $\Lambda_{i+\frac{1}{2}}$;

3. $\bar{\mathbf{F}}_{i+\frac{1}{2}}$;

4. the LLF bar state.

The bar state is

$$\boxed{\bar{\mathbf{U}}_{i+\frac{1}{2}} = \frac{\mathbf{U}_i + \mathbf{U}_{i+1}}{2} - \frac{\mathbf{f}(\mathbf{U}_{i+1}) - \mathbf{f}(\mathbf{U}_i)}{2\Lambda_{i+\frac{1}{2}}}.}$$

Write

$$\bar{\mathbf{U}}_{i+\frac{1}{2}} = \begin{bmatrix} \bar{h}_{i+\frac{1}{2}} \\ \bar{q}_{i+\frac{1}{2}} \end{bmatrix}, \qquad \bar{v}_{i+\frac{1}{2}} = \frac{\bar{q}_{i+\frac{1}{2}}}{\bar{h}_{i+\frac{1}{2}}}.$$

8 8. Step 2 — NN predicts the raw subgrid flux

Feed

$$(\mathbf{U}_{i-1}, \mathbf{U}_i, \mathbf{U}_{i+1}, \mathbf{U}_{i+2})$$

to the trained NN:

$$\boxed{\widetilde{\mathbf{G}}_{i+\frac{1}{2}} = \widetilde{\mathcal{G}}(\mathbf{U}_{i-1}, \mathbf{U}_i, \mathbf{U}_{i+1}, \mathbf{U}_{i+2}; \omega).}$$

At this point, $\widetilde{\mathbf{G}}$ is only a *raw prediction*. If used directly,

$$\bar{\mathbf{F}} + \widetilde{\mathbf{G}}$$

could produce spurious oscillations or negative water heights.

This is why MCL is inserted.

9 9. Step 3 — MCL constructs a safe correction

The paper defines corrected intermediate bar states by

$$\boxed{\bar{\mathbf{U}}^{*,\pm}_{i+\frac{1}{2}} = \bar{\mathbf{U}}_{i+\frac{1}{2}} \pm \frac{\widetilde{\mathbf{G}}^{*}_{i+\frac{1}{2}}}{\Lambda_{i+\frac{1}{2}}}.}$$

MCL chooses $\widetilde{\mathbf{G}}^{*}$ so that these intermediate states satisfy the required local bounds.

Important

MCL does not independently clip the two NN output components.
It first limits the height correction. Then it uses that limited height to construct compatible discharge/velocity bounds, limits a residual discharge correction, and finally reconstructs the discharge flux correction.

10 10. Step 4 — limit the height correction

Define

$$h_i^{\min} = \min(\bar{h}_{i-\frac{1}{2}}, \bar{h}_{i+\frac{1}{2}}), \qquad h_i^{\max} = \max(\bar{h}_{i-\frac{1}{2}}, \bar{h}_{i+\frac{1}{2}}),$$

and similarly

$$h_{i+1}^{\min} = \min(\bar{h}_{i+\frac{1}{2}}, \bar{h}_{i+\frac{3}{2}}), \qquad h_{i+1}^{\max} = \max(\bar{h}_{i+\frac{1}{2}}, \bar{h}_{i+\frac{3}{2}}).$$

The paper requires

$$h_i^{\min} \le \bar{h}^{*,-}_{i+\frac{1}{2}} \le h_i^{\max},$$

$$h_{i+1}^{\min} \le \bar{h}^{*,+}_{i+\frac{1}{2}} \le h_{i+1}^{\max}.$$

Since

$$\bar{h}^{*,\pm}_{i+\frac{1}{2}} = \bar{h}_{i+\frac{1}{2}} \pm \frac{\widetilde{\mathbf{G}}^{h,*}_{i+\frac{1}{2}}}{\Lambda_{i+\frac{1}{2}}},$$

these become bounds on $\widetilde{\mathbf{G}}^{h,*}$.

For the left state:

$$\Lambda_{i+\frac{1}{2}}(\bar{h}_{i+\frac{1}{2}} - h_i^{\max}) \le \widetilde{\mathbf{G}}^{h,*}_{i+\frac{1}{2}} \le \Lambda_{i+\frac{1}{2}}(\bar{h}_{i+\frac{1}{2}} - h_i^{\min}).$$

For the right state:

$$\Lambda_{i+\frac{1}{2}}(h_{i+1}^{\min} - \bar{h}_{i+\frac{1}{2}}) \le \widetilde{\mathbf{G}}^{h,*}_{i+\frac{1}{2}} \le \Lambda_{i+\frac{1}{2}}(h_{i+1}^{\max} - \bar{h}_{i+\frac{1}{2}}).$$

The paper's equation (27) implements the resulting feasible interval and selects the closest bound-preserving approximation to the raw $\widetilde{\mathbf{G}}^{h}$.

The result is

$$\boxed{\widetilde{\mathbf{G}}^{h,*}_{i+\frac{1}{2}}}.$$

In particular, the constrained intermediate heights remain nonnegative.

11 11. Step 5 — decompose the discharge correction

The paper does not limit $\widetilde{\mathbf{G}}^{q}$ independently.

It defines

$$\boxed{\Delta\widetilde{\mathbf{G}}^{q}_{i+\frac{1}{2}} = \widetilde{\mathbf{G}}^{q}_{i+\frac{1}{2}} - \widetilde{\mathbf{G}}^{h,*}_{i+\frac{1}{2}}\bar{v}_{i+\frac{1}{2}}.}$$

Therefore

$$\boxed{\widetilde{\mathbf{G}}^{q} = \widetilde{\mathbf{G}}^{h,*}\bar{v} + \Delta\widetilde{\mathbf{G}}^{q}.}$$

After limiting,

$$\boxed{\widetilde{\mathbf{G}}^{q,*} = \widetilde{\mathbf{G}}^{h,*}\bar{v} + \Delta\widetilde{\mathbf{G}}^{q,*}.}$$

This is why the height and discharge corrections remain compatible.

12 12. Step 6 — limit the residual discharge correction

Define local velocity bounds:

$$v_i^{\min} = \min(\bar{v}_{i-\frac{1}{2}}, \bar{v}_{i+\frac{1}{2}}), \qquad v_i^{\max} = \max(\bar{v}_{i-\frac{1}{2}}, \bar{v}_{i+\frac{1}{2}}),$$

with analogous definitions for cell $i+1$.

The corrected states must satisfy

$$\bar{h}^{*,-}_{i+\frac{1}{2}} v_i^{\min} \le \bar{q}^{*,-}_{i+\frac{1}{2}} \le \bar{h}^{*,-}_{i+\frac{1}{2}} v_i^{\max},$$

and

$$\bar{h}^{*,+}_{i+\frac{1}{2}} v_{i+1}^{\min} \le \bar{q}^{*,+}_{i+\frac{1}{2}} \le \bar{h}^{*,+}_{i+\frac{1}{2}} v_{i+1}^{\max}.$$

Using

$$\bar{q}^{*,\pm} = \bar{h}^{*,\pm}\bar{v} \pm \frac{\Delta\widetilde{\mathbf{G}}^{q,*}}{\Lambda},$$

the paper derives the bounds in its equations (29)–(30).

Equation (31) limits the raw residual

$$\Delta\widetilde{\mathbf{G}}^{q}$$

to the feasible interval, producing

$$\boxed{\Delta\widetilde{\mathbf{G}}^{q,*}}.$$

Then reconstruct:

$$\boxed{\widetilde{\mathbf{G}}^{q,*} = \widetilde{\mathbf{G}}^{h,*}\bar{v} + \Delta\widetilde{\mathbf{G}}^{q,*}.}$$

Finally,

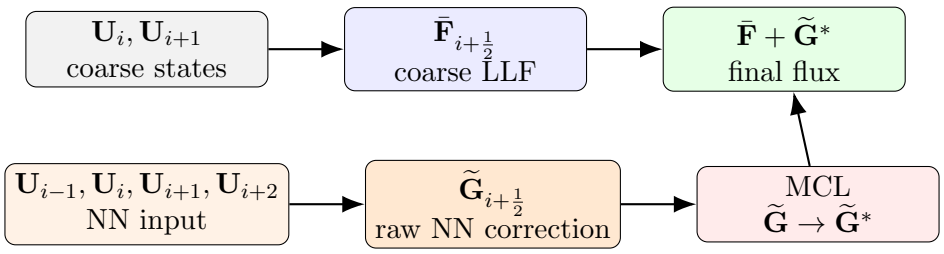
$$\boxed{\widetilde{\mathbf{G}}^{*}_{i+\frac{1}{2}} = \begin{bmatrix} \widetilde{\mathbf{G}}^{h,*}_{i+\frac{1}{2}} \\ \widetilde{\mathbf{G}}^{q,*}_{i+\frac{1}{2}} \end{bmatrix}.}$$

13 13. Step 7 — construct the final flux

The actual flux used by the coarse solver is

$$\boxed{\mathbf{F}^{\text{coarse}}_{i+\frac{1}{2}} = \bar{\mathbf{F}}_{i+\frac{1}{2}} + \widetilde{\mathbf{G}}^{*}_{i+\frac{1}{2}}.}$$

Visualized:



14 14. Step 8 — update the coarse PDE

The coarse evolution equation is

$$\boxed{\frac{d\mathbf{U}_i}{dt} = \frac{(\bar{\mathbf{F}}_{i-\frac{1}{2}} + \widetilde{\mathbf{G}}^*_{i-\frac{1}{2}}) - (\bar{\mathbf{F}}_{i+\frac{1}{2}} + \widetilde{\mathbf{G}}^*_{i+\frac{1}{2}})}{\Delta X}.}$$

Equivalently,

$$\boxed{\frac{d\mathbf{U}_i}{dt} = \frac{\mathbf{F}^{\text{coarse}}_{i-\frac{1}{2}} - \mathbf{F}^{\text{coarse}}_{i+\frac{1}{2}}}{\Delta X}.}$$

This is the actual equation that evolves the coarse solution.

15 15. Why the paper writes the bar-state form

The same equation can be written as

$$\boxed{\frac{d\mathbf{U}_i}{dt} = \frac{1}{\Delta X}\left[\Lambda_{i-\frac{1}{2}}(\bar{\mathbf{U}}^{*,+}_{i-\frac{1}{2}} - \mathbf{U}_i) + \Lambda_{i+\frac{1}{2}}(\bar{\mathbf{U}}^{*,-}_{i+\frac{1}{2}} - \mathbf{U}_i)\right].}$$

This is not a second unrelated update rule. It is an equivalent rewriting of the flux form.

The reason the paper introduces it is that MCL controls the intermediate states $\bar{\mathbf{U}}^{*,\pm}$ directly. The bar-state form therefore makes the positivity/invariant-domain proof transparent.

16 16. Where the convex combination comes from

Take one forward Euler stage:

$$\mathbf{U}^{\text{FE}}_i = \mathbf{U}_i + \frac{\Delta T}{\Delta X}\left[\Lambda_{i-\frac{1}{2}}(\bar{\mathbf{U}}^{*,+}_{i-\frac{1}{2}} - \mathbf{U}_i) + \Lambda_{i+\frac{1}{2}}(\bar{\mathbf{U}}^{*,-}_{i+\frac{1}{2}} - \mathbf{U}_i)\right].$$

Collect the three states:

$$\boxed{\mathbf{U}^{\text{FE}}_i = \alpha\mathbf{U}_i + \beta\bar{\mathbf{U}}^{*,+}_{i-\frac{1}{2}} + \gamma\bar{\mathbf{U}}^{*,-}_{i+\frac{1}{2}},}$$

where

$$\beta = \frac{\Delta T}{\Delta X}\Lambda_{i-\frac{1}{2}}, \qquad \gamma = \frac{\Delta T}{\Delta X}\Lambda_{i+\frac{1}{2}},$$

and

$$\alpha = 1 - \frac{\Delta T}{\Delta X}(\Lambda_{i-\frac{1}{2}} + \Lambda_{i+\frac{1}{2}}).$$

Therefore

$$\alpha + \beta + \gamma = 1.$$

If

$$\boxed{\frac{\Delta T}{\Delta X}(\Lambda_{i-\frac{1}{2}} + \Lambda_{i+\frac{1}{2}}) \le 1,}$$

then

$$\alpha, \beta, \gamma \ge 0.$$

This is the CFL condition.

17 17. Why this gives $h \geq 0$

The physical admissible set is

$$\mathcal{A} = \left\{ \begin{bmatrix} h \\ q \end{bmatrix} : h \geq 0 \right\}.$$

The ordinary LLF bar states are admissible when the neighboring coarse states are admissible.

MCL constructs the limited bar states so that

$$\bar{\mathbf{U}}^{*,+}_{i-\frac{1}{2}} \in \mathcal{A}, \qquad \bar{\mathbf{U}}^{*,-}_{i+\frac{1}{2}} \in \mathcal{A}.$$

Under the CFL condition,

$$\mathbf{U}_i^{\mathrm{FE}} = \alpha \mathbf{U}_i + \beta \bar{\mathbf{U}}^{*,+}_{i-\frac{1}{2}} + \gamma \bar{\mathbf{U}}^{*,-}_{i+\frac{1}{2}}$$

is a convex combination.

Because $\mathcal{A}$ is convex,

$$\boxed{\mathbf{U}_i^{\mathrm{FE}} \in \mathcal{A}}$$

and hence

$$\boxed{h_i^{\mathrm{FE}} \geq 0.}$$

So the positivity guarantee comes from the combination of:

$$\boxed{\text{MCL-admissible intermediate states} + \text{CFL condition} + \text{convexity.}}$$

18 18. Step 9 — Heun time integration

The paper uses Heun's explicit second-order SSP Runge–Kutta method.

Let $\mathcal{L}(\mathbf{U})$ denote the complete spatial operator obtained by recomputing LLF fluxes, NN corrections, MCL corrections, and final fluxes.

The first stage is

$$\boxed{\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta T\, \mathcal{L}(\mathbf{U}^n).}$$

The second stage is

$$\boxed{\mathbf{U}^{n+1} = \frac{1}{2}\mathbf{U}^n + \frac{1}{2}\left[\mathbf{U}^{(1)} + \Delta T\, \mathcal{L}(\mathbf{U}^{(1)})\right].}$$

Thus the whole LLF + NN + MCL procedure is evaluated again for the second stage.

The SSP structure is what lets the positivity/invariant-domain argument for the forward Euler building block extend to the fully discrete Heun method under the required timestep restriction.

19 19. Complete inference algorithm

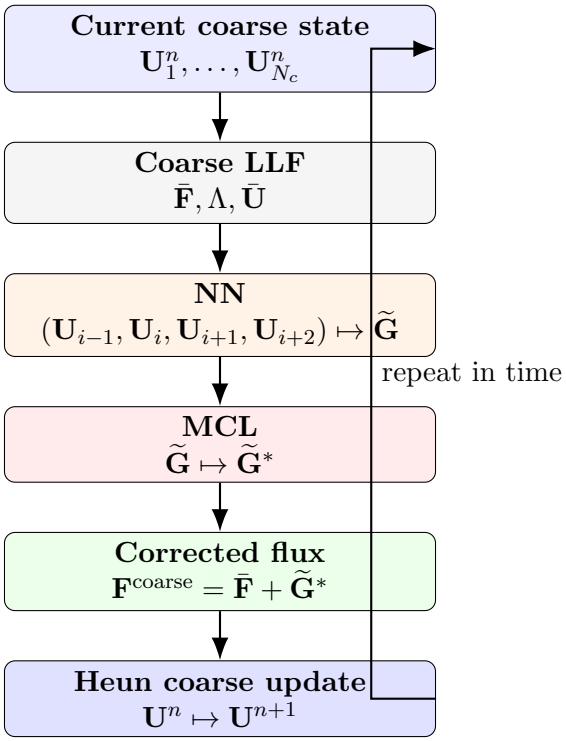
Inference algorithm

Given: trained NN $\widetilde{\mathcal{G}}$ and initial coarse state $\mathbf{U}^0$.
For every coarse timestep:

1. For every interface $i+\frac{1}{2}$, obtain $\mathbf{U}_i, \mathbf{U}_{i+1}$.
2. Compute $\Lambda_{i+\frac{1}{2}}$.
3. Compute the coarse LLF flux $\bar{\mathbf{F}}_{i+\frac{1}{2}}$.
4. Compute the LLF bar state $\bar{\mathbf{U}}_{i+\frac{1}{2}}$.
5. Feed $(\mathbf{U}_{i-1}, \mathbf{U}_i, \mathbf{U}_{i+1}, \mathbf{U}_{i+2})$ to the trained NN.
6. Obtain the raw correction $\widetilde{\mathbf{G}}_{i+\frac{1}{2}}$.
7. Limit $\widetilde{\mathbf{G}}^h$ using the height bounds.
8. Form $\Delta\widetilde{\mathbf{G}}^q = \widetilde{\mathbf{G}}^q - \widetilde{\mathbf{G}}^{h,*}\bar{v}$.
9. Limit $\Delta\widetilde{\mathbf{G}}^q$ using the compatible velocity/discharge bounds.
10. Reconstruct
$$\widetilde{\mathbf{G}}^{q,*} = \widetilde{\mathbf{G}}^{h,*}\bar{v} + \Delta\widetilde{\mathbf{G}}^{q,*}.$$
11. Construct
$$\widetilde{\mathbf{G}}^{*} = [\widetilde{\mathbf{G}}^{h,*}, \widetilde{\mathbf{G}}^{q,*}]^T.$$
12. Construct
$$\mathbf{F}^{\text{coarse}} = \bar{\mathbf{F}} + \widetilde{\mathbf{G}}^{*}.$$
13. Use these fluxes in the coarse finite-volume update.
14. Repeat the spatial calculation for the second Heun stage.
15. Obtain the next coarse state $\mathbf{U}^{n+1}$.

Repeat until the desired final time is reached.

20 20. Full visual flow



21 21. What is the final prediction?

The fine solution is

$$\mathbf{u}_1(t), \ldots, \mathbf{u}_{N_f}(t).$$

Its coarse reference is

$$\boxed{\mathbf{U}_i^{\text{fine}}(t) = \frac{1}{k} \sum_{j=k(i-1)+1}^{ki} \mathbf{u}_j(t).}$$

The reduced model produces

$$\mathbf{U}_i^{\text{MCL}-\text{NN}}(t).$$

The intended comparison is

$$\boxed{\mathbf{U}_i^{\text{MCL}-\text{NN}}(t) \approx \mathbf{U}_i^{\text{fine}}(t) = C[\mathbf{u}^{\text{fine}}(t)]_i.}$$

It is *not*

$$\mathbf{U}_i^{\text{MCL}-\text{NN}}(t) \approx \mathbf{u}_j^{\text{fine}}(t).$$

Thus the NN learns the effect of unresolved fine scales on the coarse flux; it does not learn to reconstruct every unresolved fine-grid value.

22 22. Concrete parameters used in the paper

For the main numerical experiments:

$$L = 100, \qquad N_f = 2000, \qquad N_c = 100, \qquad k = 20.$$

Hence

$$\Delta x = \frac{100}{2000} = 0.05, \qquad \Delta X = 20(0.05) = 1.$$

The fine timestep is

$$\Delta t = 0.1\Delta x = 0.005.$$

The reduced model is tested with larger timesteps, including

$$\Delta T = 0.005, \qquad 0.05, \qquad 0.1.$$

The computational advantage comes from evolving only 100 coarse cells and, when stable, using a much larger timestep than the fine simulation.

23 23. Final mental picture

coarse state	$\longrightarrow$	ordinary LLF flux
		$\downarrow$
		NN predicts missing subgrid flux
		$\downarrow$
		MCL makes the correction admissible
		$\downarrow$
		LLF flux + safe NN correction
		$\downarrow$
		finite-volume/Heun update
		$\downarrow$
		next coarse state.

The roles are:

LLF	:	cheap baseline coarse physics,
NN	:	learned unresolved/subgrid correction,
MCL	:	physics-aware safety/bound preservation,
finite-volume solver	:	actually evolves the coarse state.

Therefore the reduced model is

$$\boxed{\text{coarse PDE} + \text{learned subgrid flux} + \text{MCL safety},}$$

not a neural network that directly predicts the future solution.